CMSSW angleSource=emu on cnn_nano_v3 (files: itot_emu_f03_100ev.root): residuals vs the 1.5x target reweighted to the Phase-2 angles

CMS alpha (bending) - IL2-4

CMS alpha (bending) - IL1

CMS alpha (bending) - ID1-4

CMS beta - IL2-4

CMS beta - IL1

CMS beta - ID1-4

density (unit area)

residual [deg] (ends = overflow)

SmartPixels dump 1x (PixelAV angle mix)

TARGET: dump, bias kept + 1.5x spread (PixelAV angle mix)

TARGET reweighted to the class-T Phase-2 angles (discs: at the alpha lookup key)

CMSSW emu, class T: emitted - true

CMSSW emu, class C linked: emitted - (inconsistent) TP truth

class-C route on class-T clusters (python ref., CMSSW CNN; closure)

Residual statistics - CMS alpha (core = mode-seeded Gaussian core RMS about 0)

group	curve	N	core_rms_deg	full_rms_deg	frac_gt10	note
IL2-4	SmartPixels dump 1x (PixelAV angle mix)	108222	1.27	3.78	1.6%	
IL2-4	TARGET: dump, bias kept + 1.5x spread (PixelAV angle mix)	108222	1.86	5.64	3.1%	
IL2-4	TARGET reweighted to the class-T Phase-2 angles (discs: at the alpha lookup key)	108222	4.48	6.95	7.2%	100% of class-T mass in dump cells
IL2-4	CMSSW emu, class T: emitted - true	1881389	4.44	6.96	7.2%	
IL2-4	CMSSW emu, class C linked: emitted - (inconsistent) TP truth	1321570	15.13	36.05	61.6%	
IL2-4	class-C route on class-T clusters (python ref., CMSSW CNN; closure)	88947	6.66	10.53	19.8%	
IL1	SmartPixels dump 1x (PixelAV angle mix)	108222	1.27	3.78	1.6%	
IL1	TARGET: dump, bias kept + 1.5x spread (PixelAV angle mix)	108222	1.86	5.64	3.1%	
IL1	TARGET reweighted to the class-T Phase-2 angles (discs: at the alpha lookup key)	108222	3.66	6.61	5.6%	100% of class-T mass in dump cells
IL1	CMSSW emu, class T: emitted - true	1130480	3.57	6.61	5.6%	
IL1	CMSSW emu, class C linked: emitted - (inconsistent) TP truth	499150	7.62	28.56	38.5%	
IL1	class-C route on class-T clusters (python ref., CMSSW CNN; closure)	53533	4.86	11.04	11.5%	
ID1-4	SmartPixels dump 1x (PixelAV angle mix)	108222	1.27	3.78	1.6%	
ID1-4	TARGET: dump, bias kept + 1.5x spread (PixelAV angle mix)	108222	1.86	5.64	3.1%	
ID1-4	TARGET reweighted to the class-T Phase-2 angles (discs: at the alpha lookup key)	108222	4.37	6.11	4.6%	100% of class-T mass in dump cells
ID1-4	CMSSW emu, class T: emitted - true	4747342	4.47	6.46	4.9%	
ID1-4	CMSSW emu, class C linked: emitted - (inconsistent) TP truth	2780251	6.71	24.03	29.9%	
ID1-4	class-C route on class-T clusters (python ref., CMSSW CNN; closure)	224322	5.11	8.74	9.4%	

Residual statistics - CMS beta (core = mode-seeded Gaussian core RMS about 0)

group	curve	N	core_rms_deg	full_rms_deg	frac_gt10	note
IL2-4	SmartPixels dump 1x (PixelAV angle mix)	108222	0.49	6.41	6.6%	
IL2-4	TARGET: dump, bias kept + 1.5x spread (PixelAV angle mix)	108222	0.72	9.44	9.4%	
IL2-4	TARGET reweighted to the class-T Phase-2 angles (discs: at the alpha lookup key)	108222	5.39	15.33	32.3%	100% of class-T mass in dump cells
IL2-4	CMSSW emu, class T: emitted - true	1881389	5.41	15.3	32.3%	
IL2-4	CMSSW emu, class C linked: emitted - (inconsistent) TP truth	1321570	28.78	44.79	67.5%	
IL2-4	class-C route on class-T clusters (python ref., CMSSW CNN; closure)	88947	9.9	22.37	46.6%	
IL1	SmartPixels dump 1x (PixelAV angle mix)	108222	0.49	6.41	6.6%	
IL1	TARGET: dump, bias kept + 1.5x spread (PixelAV angle mix)	108222	0.72	9.44	9.4%	
IL1	TARGET reweighted to the class-T Phase-2 angles (discs: at the alpha lookup key)	108222	0.85	12.06	17.9%	100% of class-T mass in dump cells
IL1	CMSSW emu, class T: emitted - true	1130480	0.83	11.94	17.9%	
IL1	CMSSW emu, class C linked: emitted - (inconsistent) TP truth	499150	2.01	45.48	47.5%	
IL1	class-C route on class-T clusters (python ref., CMSSW CNN; closure)	53533	1.65	28.62	34.3%	
ID1-4	SmartPixels dump 1x (PixelAV angle mix)	108222	0.49	6.41	6.6%	
ID1-4	TARGET: dump, bias kept + 1.5x spread (PixelAV angle mix)	108222	0.72	9.44	9.4%	
ID1-4	TARGET reweighted to the class-T Phase-2 angles (discs: at the alpha lookup key)	108222	4.88	17.83	61.6%	100% of class-T mass in dump cells
ID1-4	CMSSW emu, class T: emitted - true	4747342	19.38	18.11	62.1%	
ID1-4	CMSSW emu, class C linked: emitted - (inconsistent) TP truth	2780251	21.09	31.74	68.4%	
ID1-4	class-C route on class-T clusters (python ref., CMSSW CNN; closure)	224322	19.76	19.19	63.2%	